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Content-based encrypted speech retrieval scheme with deep hashing

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Abstract

In order to improve the limitations of manual features and poor feature semantics in the feature extraction process of existing content-based encrypted speech retrieval methods, and as well as improve retrieval accuracy and retrieval efficiency, a content-based encrypted speech retrieval scheme with deep hashing was proposed. Firstly, the original speech file is encrypted by using Henon mapping chaotic encryption to construct encrypted speech library. Secondly, adopting secondary feature extraction method to extract the spectrogram feature, and using the spectrogram as the input of the designed convolutional neural network (CNN) for model training and deep hashing feature learning, to obtain the deep hash binary code of original speech, and upload it to the deep hash index table in the cloud. In addition, the batch normalization (BN) method is introduced to improve robustness and generalization ability of the model. Finally, establish a one-to-one mapping relationship between the encrypt speech in the encrypted speech library and the hash sequence in the deep hash index table. When retrieving for speech users, the normalized Hamming distance algorithm is used for retrieve matching. The experimental results show that the deep hash binary code constructed by the proposed method has strong discriminability and robustness, and it still has high recall rate, precision rate and retrieval efficiency under various general content preserving operations.

Keywords Encrypted speech retrieval · Deep hashing · Convolutional neural network (CNN) · Spectrogram · Deep semantic feature

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1 Introduction

With the popularity of the Internet and cloud storage technology, various multimedia data has shown explosive growth. Cloud storage technology greatly saves local storage space and enables data sharing, but it also brings some challenges to data retrieval and privacy security. Among them, speech is widely concerned because of its special semantic function, and the importance and sensitivity of semantic content are reflected in applications such as conference recording, court evidence, speech order information and military secrets, therefore, cloud users want to hide this sensitive data [7]. But the cloud is not a trusted third party, the original speech is usually encrypted at the front end to ensure cloud data security. However, the encrypted speech will lose most of its features that will have a certain impact on retrieval. Hence, how to realize valid retrieval of massive encrypted speech data in the cloud has very important research value [25].

The existing content-based encrypted speech retrieval methods [8, 22, 25, 27, 32, 33, 35, 37] first analyze the speech content and extract the features, then use the obtained content features to establish an index and conduct retrieval, it adopts the idea of approximate matching (or partial matching), and obtains the retrieval results through gradual refinement, which well satisfies the retrieval requirements of the system and has become the research hotspots of many scholars at home and abroad. In the content-based encrypted speech retrieval system, most of the current researches are implemented by constructing perceptual hashing schemes [8, 22, 25, 32, 33, 35, 37], but these traditional methods use the existing manual features to perform the retrieval task, the manual features are not only computationally intensive but also subjective, they cannot accurately describe the deep semantic information and content perception information of speech data [31], which reduces the retrieval accuracy and efficiency of the retrieval system to a certain extent. Therefore, the rise of "deep learning" and "deep hashing" methods [2–6, 9, 13, 14, 17–20, 30] provides a new idea for solving this problem and has been applied to speech data processing systems by many scholars. Among them, CNN has attracted extensive attention because of its strong feature self-learning ability and representation ability. In recent years, the deep hashing method based on the CNN structure has been widely used in image classification / retrieval [13, 14, 18, 19, 30], speaker / speech recognition [2, 5, 6, 20], audio classification / retrieval [3, 4, 9, 17] and other fields, CNN is used as a feature extractor to extract the deep features of multimedia data to solve the limitations of traditional manual features, so that the whole system structure has the advantages of high precision and fast speed.

Therefore, in order to solve the problems of poor manual feature representation, low retrieval accuracy, and low retrieval efficiency in existing research, this paper designs an end-to-end deep hash learning frame based on CNN, and uses the designed CNN model as the feature extractor of speech data, and combined with the deep hashing algorithm to get the compact binary hash code.

The main innovations of this paper can be summarized as follows:

1) The deep hashing method applied in image retrieval and speech recognition is introduced into encrypted speech retrieval, so as to enhance the retrieval accuracy and efficiency, and increase the discriminability and robustness of the deep hash binary code.
2) The secondary feature extraction method is adopted to extract low-level spectrogram features, and the spectrogram is used as the CNN network input to extract deep semantic features.

3) A novel deep hash coding model is designed based on CNN to simultaneously learn semantic features and hash functions of speech data, so as to improve the feature representation ability of speech data, and generate high quality deep hash binary code.
4) By introducing the batch normalization (BN) method, the gradient's dependence on the parameters or their initial value scales is reduced, the convergence speed of the network model is accelerated, and the model becomes more robust.

The other parts of this paper are organized as follows: Section 2 summarizes the relevant research work in this field. Section 3 shows the theoretical knowledge involved in this study. Section 4 elaborates on the encrypted speech retrieval method proposed in this paper. Section 5 gives the relevant experimental test results. Finally, we conclude our work in Section 6.

2 Related work

In recent years, the content-based encrypted speech retrieval method has received extensive attention, and many scholars have proposed different encrypted speech retrieval schemes. Mainly including: retrieval method based on perceptual hashing [8, 22, 25, 32, 33, 35, 37], retrieval method based on content privacy protection [32], retrieval method based on searchable encryption [7, 12, 15], retrieval method based on local sensitive hashing [23] and retrieval method based on deep learning [7, 34, 36], etc., among them, the most widely studied is the encrypted speech retrieval method based on perceptual hashing. Wang and Hao [25] proposed a perceptual hashing encrypted speech retrieval scheme based on time-frequency transform, the speech signal is divided into time domain and frequency domain to construct perceptual hash sequences according to the short-term energy, it realizes the retrieval of encrypted speech under the condition of "no download, no decryption". Zhao and He [37] proposed the use of multifractal characteristic of speech and piecewise aggregate approximation to generate perceptual hash sequences with good discriminability and robustness, and then achieve fast speech retrieval of encrypted speech. Zhang et al. [32] proposed to extract the short-term cross-correlation coefficients of the speech and form a coefficient matrix as perceptual hash feature, and use scrambling encryption to extract perceptual hash digest directly from the encrypted speech, making the system more secure and efficient. He and Zhao [8] proposed an encrypted speech retrieval system based on the syllable-level perceptual hashing, it adopts the posterior probability of the acoustic model to get the perceptual hash digest, which implements function of retrieving speech segment and spoken term on the encrypted speech database. Zhang et al. [35] proposed an encrypted speech retrieval method based on Chirp-Z transformation and perceptual hash secondary feature extraction, the perceptual hash digest is extracted through Chirp-Z transformation and sparse matrix, and the speech file is encrypted according to the m sequence, which has the well retrieval effect on noisy speech. Zhang et al. [33] proposed an efficient retrieval algorithm of encrypted speech based on inverse fast Fourier transform and measurement matrix, this algorithm improves the security of plaintext data and has good retrieval performance and discrimination, but its robustness against MP3 compression attacks needs to be further improved. The above-mentioned perceptual hashing methods can produce perceptual hash codes with good discriminability and robustness to a certain extent, but these methods are based on the traditional manual features, so their semantics are not high, which affects the accuracy and efficiency of the retrieval system.

In view of this, the excellent performance of deep learning method in various fields is worth of reference, deep learning and deep hashing methods have achieved great success in the fields of image classification and retrieval [13, 14, 18, 19, 30], speaker / speech recognition [2, 5, 6, 20], speech classification and retrieval [4, 7, 17, 34, 36], etc. For example, Glackin et al. [7] proposed a cloud-based encrypted speech search strategy, which established an acoustic model based on CNN on the client, encoded the speech data into character marks, and encrypted the speech files with AES and uploaded them to the cloud, the server performs the search task of speech content, which effectively ensuring the security of private data. Qin et al. [18] proposed a new deep hash learning framework based on deep residual network, which uses multi-scale receiving field convolutional neural network to extract features of the segmented images and learn hash coding. Li et al. [14] proposed a deep hashing image retrieval method, it optimizes the image hash coding by using two times of feature binarization, so as to maximize semantic similarity and get the most similar retrieval results with the query. Qin et al. [19] used an optimized AlexNet network to extract image feature descriptors and generate high-quality hash codes, it can ensure that similar images are kept at the top of the result list and improve the retrieval performance. Fan et al. [6] proposed a novel deep hashing method, which performs feature learning and hash code learning in an end-to-end architecture, it improves the performance of speaker recognition and retrieval tasks. Bartz et al. [2] proposed an automatic language identification system, it adopts convolutional recurrent neural network to operate on the speech spectrum, the trained model is suitable all kinds of noisy environments and has great recognition accuracy. Dhiraj et al. [4] proposed an audio feature extraction and visualization method based on deep learning, it uses a CNN auto-encoder based on visual geometry group network (VGGNet) to classify music genres. Patil and Nemade [17] combined fuzzy probability and probabilistic neural network (PNN) features, proposed a method of combining machine learning and neural network – FPNN to perform preprocessing, segmentation, feature extraction, classification and retrieval tasks for audio, and the results have good classification accuracy. Zhang et al. [36] proposed an encrypted speech retrieval method based on CNN-BiLSTM and deep hashing, using the CNN-BiLSTM fusion model to generate the deep hash coding of speech data, and using 4D hyperchaotic encryption algorithm to encrypt the original speech, the whole system has a high recall rate, precision rate and security. Zhang et al. [34] proposed a deep polynomial network (DPN), which can be used as an acoustic model to be applied to encrypted speech, it uses homomorphic encryption technology to encrypt the acoustic model data, at the same time, DPN can perform frame-level prediction on encrypted speech and return it in encrypted form, which has high security.

Due to the high dimension of speech data, the low dimension feature representation is crucial. At present, many speech feature extraction methods have also been continuously studied. The traditional most representative ones are feature extraction based on perceptual characteristics [8, 22, 25, 32, 33, 35, 37], feature extraction based on Mel frequency cepstral coefficients (MFCC) [9, 28], feature extraction based on power normalized cepstral coefficient (PNCC) [10], feature extraction based on entropy [24], feature extraction based on hybrid features [21], etc., to some extent, these traditional feature extraction methods can effectively represent the speech data in low dimension, but it can only express low-level perceptual features, the advanced semantic information of speech data cannot be fully expressed. Therefore, deep neural network (DNN) is used to extract deep semantic features of speech due to its powerful feature learning ability. Fan et al. [6] proposed a deep additive margin hash (DAMH) method, it uses an improved CNN structure to extract speaker speech features. Kim and Pardo [11] proposed using Log-Mel spectrum as input and CNN as feature extractor to extract audio feature vectors. Nayyar et al.

[16] trained two neural network models: fully convolutional neural network (FCNN) and convolutional recurrent neural network (CRNN), using Mel-spectrum as input to learn speech feature representation. Xu et al. [29] applied a CRNN with learnable gated linear units (GLUs) to the Mel spectrum, to learn the most relevant speech features.

To sum up, aiming at the shortcomings of feature extraction methods and low system performance of existing methods, and the advantages of deep hashing method in other fields such as higher retrieval accuracy and faster retrieval speed, we proposed a content-based encrypted speech retrieval scheme with deep hashing. In this way, we can improve the semantic representation of speech features and the performance of the retrieval system, so as to achieve efficient and accurate retrieval of encrypted speech data in cloud.

3 Related theory analysis

3.1 Spectrogram features

The spectrogram [6] is a spectrum analysis view of speech, it combines the characteristics of spectrum and time domain waveform. It can express three-dimensional information through a two-dimensional plane, which clearly reflects the change of the speech spectrum with time. Figure 1 shows the processing flow of spectrogram feature.

The specific generation steps can be summarized as:

Step 1: Resampling format conversion. Open the speech file $x(n)$, and convert the obtained speech parameters in string format into integer format.

Step 2: Amplitude normalization. The wave amplitude of speech signal is normalized, the calculation formula as shown in Eq. (1).

$$\hat{x}(n) = x(n) \times 1.0/\max(abs(x(n))) \tag{1}$$

where $\hat{x}(n)$ is the speech signal after normalization processing, $x(n)$ is the speech signal without normalization processing.

Step 3: Framing and adding window. Framing the speech signal, generally the frame length is 20-30 ms, and set the window function and frame overlap. The window function adopts Hanning window, and frame overlap is set to 1/3 of the points per frame, the Hanning window function as shown in Eq. (2).

$$w(n) = \begin{cases} 0.5\left(1 - \cos\frac{2n\pi}{M-1}\right) & (0 \le n \le M-1) \\ 0 & (n < 0 \text{ or } n > M) \end{cases} \tag{2}$$

The calculation formula for adding window as shown in Eq. (3).

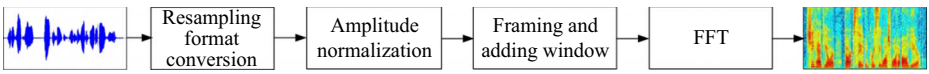


Fig. 1 Spectrogram feature processing flow

$$x_i(n) = w(n) \times \hat{x}(iT + n), \quad 1 \le n \le N \tag{3}$$

where $w(n)$ is the Hanning window function, M is the length of the Hanning window, $x_i(n)$ is the i-th frame speech signal obtained by adding window division, T is the frame shift length, and N represents the total frame length.

Step 4: Fast Fourier Transform (FFT). Performing FFT on the speech signal after framing and adding window, and converting the time domain information into frequency domain information, the calculation formula as shown in Eq. (4).

$$X_i(k) = \sum_{n=0}^{N-1} x_i(n) \exp\left(-j\frac{2\pi kn}{N}\right), \quad 0 \le k \le N-1 \tag{4}$$

where $X_i(k)$ is the speech data after FFT transformation, k is the point number.

Step 5: Generate spectrogram features. The frequency domain information after the FFT processing is spliced according to time series to form spectrogram feature $S(n)$.

3.2 Deep hashing method

With the continuous improvement of user retrieval requirements, the approximate nearest neighbor (ANN) search method [14] has been widely used in retrieval tasks in recent years, as a representative method, hashing has received widespread attention. The goal of the hash method is to represent samples as a hash sequence, so that similar samples have similar binary codes, and dissimilar samples have different binary codes [37]. However, the basis for obtaining high-quality hash code is to have better feature representation, while traditional manual features cannot dig into the deep semantic information of speech, so the introduction of deep learning method is crucial. Figure 2 shows the structural schematic of the deep hashing method.

As shown in Fig. 2, the existing deep hashing methods [6, 14, 19, 30] can be summarized as follows: deep neural network model learning feature representation + activation function (sigmoid/tanh/relu) limiting output range + constructing different hash functions mapping + designing different loss functions/network structures. Its construction principle is mainly to use the deep neural network model to mine the deep semantic features of input data, then use hash function constructed by the model to map the deep semantic features, so as to obtain the more meaningful feature hash code to improve the retrieval performance of the system. However, when constructing deep hash code, because the goal of hash method is to get a binary code, therefore in the process of optimization, we will encounter the constraints of discrete values, so

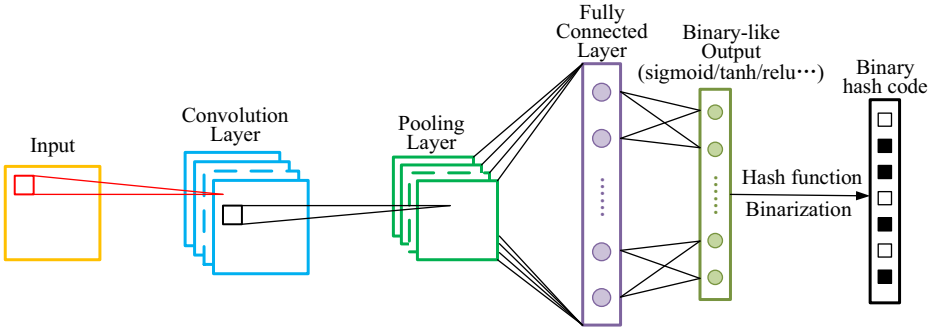


Fig. 2 Structural schematic of deep hashing method

the gradient based method cannot be used to optimize the objective function. Therefore, the deep hashing method uses a looser constraint instead of requiring the "binary code" to be binary, but only within a certain range, that is to say, limiting the output range at the hash-like layer through the activation function, after the optimization, the relaxed "binary code" is quantified by the constructed hash function to get the true binary code.

3.3 Chaos encryption based on Henon map

Chaos is a complex dynamic behavior, it has no obvious period and symmetry, and the complex internal structure is a new form of nonlinear system. The chaotic signal has the characteristics of aperiodicity, continuous broadband spectrum and noise like, which makes it have good concealment, and the chaotic signal generated by the nonlinear algorithm can provide a large number of keys under the encryption algorithm, which is used to generate the sequence cipher and has high security.

As a representative of two-dimensional chaotic system, Henon map [1] is widely used in multimedia data encryption because of its good randomness and encryption effect. The mapped nonlinear dynamic system is:

$$\begin{cases} \mu_{n+1} = 1 - \alpha\mu_n^2 + \nu_n \\ \nu_{n+1} = \beta\mu_n \end{cases} \tag{5}$$

where α, β are system variables, when $\alpha = 1.4$ and $\beta = 0.3$, the system is in chaos. At this time, the maximum Lyapunov exponent is equal to 0.418.

It can be seen from Eq. (5), the Henon mapping is to determine the iterative equation by two variables u and v simultaneously, it is more complicated than simply combining two independent one-dimensional chaotic equations, and it has better performance and is suitable as a pseudo-random sequence generator for multimedia data encryption. Therefore, this paper uses the Henon mapping equation of Eq. (5) to generate the Henon chaotic encryption sequence, and then encrypts the original speech files.

4 The proposed method

The proposed method uses the CNN-based model to extract the deep semantic features, and then construct deep hashing sequence through the learned hash scheme, so as to obtain the compact binary code representations of speech data, which lays a solid foundation for subsequent speech retrieval tasks.

4.1 System retrieval model

Figure 3 shows the encrypted speech retrieval model based on deep hashing proposed in this paper. From Fig. 3, we can see that the retrieval system mainly consists of three modules: encrypted speech library building module, deep hashing and deep hash index table construction module and query speech retrieval module. In the process of constructing deep hashing and deep hash index table, a new DNN model suitable for speech data hash coding based on the CNN structure is designed in this paper, adopting the second feature extraction method [35] and using the spectrogram features as network input to perform semantic feature and hash

function learning at the same time, so as to get the deep hash coding with rich semantic information. Besides, in order to speed up the convergence of the network model and reduce the risk of over-fitting, BN algorithm is added to the model training, so that the trained model has better robustness. In the proposed scheme, the introduction of deep hashing method is helpful to obtain the more distinguishable and robust deep hash binary code, which can further improve the query accuracy and efficiency, and effectively solve the problems of manual feature flaws in the feature extraction process and low retrieval precision and retrieval efficiency of the traditional encrypted speech retrieval system based on perceptual hashing.

4.2 Construction of encrypted speech library

This paper uses the Henon mapping chaos encryption algorithm of Section 3.3 to encrypt the original speech file, the encryption process is as follows:

Step 1: Convert original speech files X_n into matrix form and record them as M_n.
Step 2: Determine the initial encryption key $[\mu_0, \nu_0]$, and generate the Henon chaotic encryption sequence $\{K_\mu\}$ and $\{K_\nu\}$, according to the Eq. (5) Henon mapping equation.
Step 3: Multiply the $\{K_\mu\}$ and $\{K_\nu\}$ chaotic sequence values by 256, take the integer part as the new sequence value, and form the matrix C_μ and C_ν with the same size as the speech file matrix M_n.
Step 4: Perform bitwise XOR operations between original speech file matrix M_n and matrix C_μ and C_ν to produce Henon encrypted speech files E_n.
Step 5: After the above encryption steps, the encrypted speech files En are uploaded to the cloud, the encrypted speech library is successfully constructed.

4.3 Construction of the deep hashing scheme

4.3.1 Generation model of deep hash binary code

Figure 4 shows the schematic diagram of CNN-based deep hashing. CNN has powerful feature learning ability, which can learn deep features through convolutional layers and pooling layers.

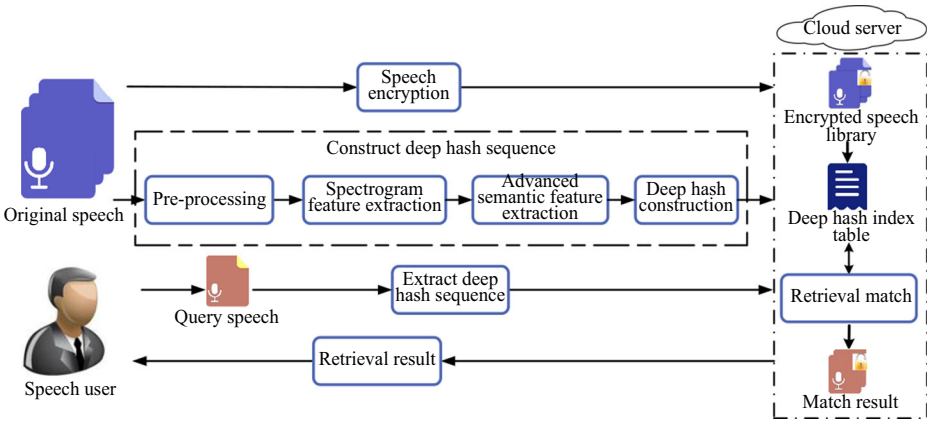


Fig. 3 The system model of the proposed method

The proposed method takes advantage of CNN to design a novel CNN model that can efficiently extract the semantic features of speech.

It can be seen from Fig. 4, the proposed CNN model contains 4 convolutional layers, 4 pooling layers and 3 fully connected layers. The first convolutional layer contains 32 convolution filters, the kernel size is 3×3 and the data input size is 224×224 of 3 channels; then there is a maximum pooling layer, the filter size is 2×2; the second convolutional layer contains 64 convolution filters with the kernel size of 3×3; after that, a maximum pooling layer with the filter size of 2×2 is still connected; after the second pooling layer, add a batch normalization layer to speed up network fitting as much as possible. Next, the third convolutional layer still sets the kernel size to 3×3, but sets the number of convolution filters to 128; later connect the maximum pooling layer with the same settings as before; the fourth convolutional layer and the maximum pooling layer retain the parameter settings of the third convolutional and pooling layer; after the fourth pooling layer, the batch normalization layer is still added. Then connect a FC fully connected layer to flatten the extracted features; after the FC layer is the hash layer, it can get the hash binary code; After the hash layer, set an output layer whose number of nodes is number of speech sample categories, and use the softmax function, the output layer is added because the activity of neurons in the hash layer will be adjusted by the later output layer when the model is trained, and the output layer has rich category semantic information, so it is helpful for the hash layer to obtain rich semantic feature representation. Table 1 Shows the specific settings of the model parameters.

In order to prevent over-fitting, dropout is applied to the fully connected layer, and set the discard rate to 0.2, that is, randomly discard 20% of neuron nodes does not participate in model training. The nonlinear activation function in each convolutional layer and fully connected layer in this model uses the Tanh function. In addition, a too high learning rate will cause the model not to converge, and a too low learning rate will cause the model to converge particularly slowly or unable to learn, according to the experience selection and test experiment, the learning rate of the model is set to 0.01. At the same time, in the later stage of model iterative optimization, gradually reducing the value of the learning rate will help the model to converge and make it easier access to optimal solution. Therefore, according to the experimental test, setting the learning rate decay to 1×10^{-6}. The loss function uses binary_crossentropy, and the optimizer algorithm is stochastic gradient descent (SGD). Finally, set the training batch (epoch) to 40, and batch size to 64 for model training, a deep hash coding model suitable for speech data is obtained.

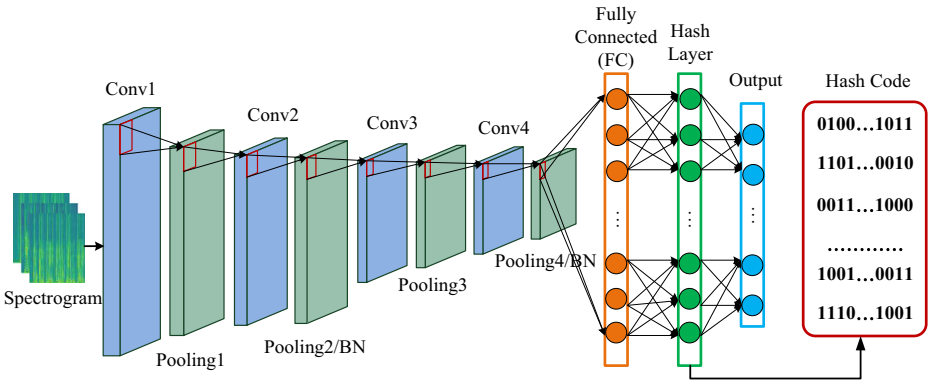


Fig. 4 Schematic diagram of CNN-based deep hashing

Table 1 Parameter setting table of deep hash coding model

Layer	Configuration
Conv1	Conv2D(filters=32, kernel_size=(3, 3), activation = 'tanh', input_shape=(224, 224, 3))
MaxPooling1	MaxPooling2D(pool_size=(2, 2))
Conv2	Conv2D(filters=64, kernel_size=(3, 3), activation = 'tanh')
MaxPooling2	MaxPooling2D(pool_size=(2, 2))
BN	BatchNormalization (axis=−1)
Conv3	Conv2D(filters=128, kernel_size=(3, 3), activation = 'tanh')
MaxPooling3	MaxPooling2D(pool_size=(2, 2))
Conv4	Conv2D(filters=128, kernel_size=(3, 3), activation = 'tanh')
MaxPooling4	MaxPooling2D(pool_size=(2, 2))
BN	BatchNormalization (axis=−1)
FC	Flatten()
Dropout	Dropout(0.2)
Hash Layer	Dense(384, activation = 'tanh')
Output	Dense(10, activation = 'softmax')

4.3.2 Hash function learning and deep hash binary code construction

Given a training set of N sample size $X = \{x_1, x_2, \ldots, x_N\} \in R^{d \times N}$, where $x_i \in R^d$ represents the i-th sample in the training set X. The purpose of hash learning is to learn a nonlinear function $h(\cdot)$, which can convert the data in the original multi-dimensional space Ω into the binary representation of K bits (ie: $\Omega \rightarrow \{0, 1\}^{K \times N}$), and ensure that the binary codes after the conversion of similar data in the original space Ω are still similar, and the binary codes after different data conversion are quite different. For the deep hashing construction method in the proposed method, the above conversion process can be decomposed into: $f(X) = h(\mu(X)) \in \{0, 1\}^{K \times N}$. Among them, $\mu(X)$ is the process of extracting deep semantic features of speech data using the designed CNN model; $h(\cdot)$ represents the process of converting the extracted semantic features into binary codes. The mathematical expression of $\mu(X)$ as shown in Eq. (6):

$$\mu(X) = W_b^T \varphi(X, W_a) + V_b \tag{6}$$

where W_a is the weight of the convolutional layer and pooling layer in feature learning, $\varphi(\cdot)$ represents the point multiplication of the weight matrix of the convolution and pooling layer and the input data, W_b represents the weight matrix of hash layer, and V_b represents the hash layer offset.

The sign function $sign(\cdot)$ is adopted to map semantic features to generate the binary code of speech data. Specifically, the process of mapping the sample x_i into a K-bit binary code is:

$$b_i = h(\mu(x_i)) = \text{sign}\left(\left(W_b^T \varphi(x_i, W_a) + V_b\right) - I_{mean}\right) \tag{7}$$

where, I_{mean} is the mean value of the extracted semantic feature vectors.

Combining with Eqs. (6) and (7) can be integrated into:

$$b_i = sign(\mu(x_i) - I_{mean}) \tag{8}$$

Let $\mu(x_i) - I_{mean} = \phi$, then Eq. (8) can be expressed as:

$$b_i = h(\mu(x_i)) = \text{sign}(\phi) = \begin{cases} 1, & \phi \geq 0 \\ 0, & otherwise \end{cases} \tag{9}$$

Through Eq. (9), the deep hash binary code representation of the spectrogram input sample as shown in Fig. 4 can be obtained, that is, the construction of the deep hashing sequence of the speech data is completed.

According to the hash function learning and the related principles of deep hash binary code construction in the model, the process of deep hashing sequence construction and cloud deep hash index table establishment can be summarized as follows:

Step 1: Feature extraction of primary spectrogram. For the original speech files $X = \{x_1, x_2, \ldots, x_N\}$ in the training set, perform pre-processing operations such as pre-emphasis, framing, adding window, etc., and then use the method described in Section 3.1 to obtain the spectrogram feature $S(n)$ of each speech file.

Step 2: Network model training. The obtained spectrogram features $\{S_1, S_2, \ldots, S_n\}$ as the input of the network model designed in Section 4.3.1, after the experiments and parameter adjustments are carried out continuously to obtain the optimal model for speech data input.

Step 3: Advanced semantic feature extraction and deep hashing construction. Still using the spectrogram features $\{S_1, S_2, \ldots, S_n\}$ as the input of the trained network model, the deep semantic feature extraction is performed by Eq. (6), and through the learned hash function to construct the deep hashing sequence of semantic features, thus get the deep hash binary code of each speech file $\{b_1, b_2, \ldots, b_n\}$.

Step 4: Structuring deep hash index table. Establish one-to-one mapping relationship between deep hash binary codes $\{b_1, b_2, \ldots, b_n\}$ of all speech files and the corresponding encrypted speech files $\{E_1, E_2, \ldots, E_n\}$, and upload them to the cloud to obtain final deep hash index table.

4.4 Speech retrieval and speech decryption

When users need to search, first input the query speech, then use the deep hashing sequence construction scheme that used in building the deep hash index table to extract the deep hash binary code of the query speech, finally the deep hash binary code of the query speech and the deep hashing sequence in the deep hash index table are searched and matched using the normalized Hamming distance algorithm. After successful retrieval, the corresponding encrypted speech file in the encrypted speech library is decrypted and fed back to the speech query users. The process is as follows:

Step 1: Query speech preprocessing. Input the query speech x_q, perform pre-processing operations such as pre-emphasis, framing, adding window, etc., so as to get the reprocessed speech signal x_q'.

Step 2: Spectrogram feature extraction. Extract the low-level spectrogram feature of the speech signal x_q' and record it as S_q.

Step 3: Deep hash binary code construction. Put the spectrogram feature S_q into the trained network model in Section 4.3.1, and extract its deep semantic feature according to the method described in Section 4.3.2, then the deep hash binary code b_q is gained in hash layer through the deep hashing construction of Eq. (9).

Step 4: Hamming distance matching. Set the distance threshold τ, calculate the normalized Hamming distance $D(b_q, b_x)$ between the deep hash binary code b_q of the query

speech and the hash sequence and b_x in the deep hash index table. Eq. (10) shows the normalized Hamming distance formula:

$$D(b_x, b_q) = \frac{1}{N}\sum_{i=1}^{N}\left(\left|b_x(i) - b_q(i)\right|\right) = \frac{1}{N}\sum_{i=1}^{N} b_x(i) \oplus b_q(i), \quad i = 1, 2, , N \tag{10}$$

where N refers to the length of hash code, $\oplus$ represents XOR operation.

If $D(b_q, b_x) > \tau$, it means that the perception content of the two speeches is different; if $D(b_q,b_x) \leq \tau$, it means that the perception content is same or similar, so the corresponding encrypted speech file matching successfully is decrypted and returned to the speech query users.

Decryption algorithm is the inverse operation of encryption algorithm, and the main process is as follows:

Step 1: Open returned encrypted speech file E_q and convert it into matrix form E_q'.

Step 2: According to Henon mapping equation of Eq. (5), input the initial key $[\mu_0,\nu_0]$ to generate the Henon chaotic decryption sequence $\{K_\mu\}$ and $\{K_\nu\}$.

Step 3: Multiply the $\{K_\mu\}$ and $\{K_\nu\}$ chaotic sequence value by 256, and take the integer part as the new sequence value, and then form the matrix C_μ and C_ν with the same size as E_q'.

Step 4: Implement bitwise XOR operation on the matrices C_μ and C_ν and the encrypted speech matrix E_q' successively. After processing, obtain the Henon decrypted speech file x_q, which is the retrieval result of the query speech.

5 Experimental results and performance analysis

This paper uses the speeches in the THCHS-30 [26] database to evaluate the proposed method, the speech sampling frequency is 16 kHz, the sampling size is 16 bits, and it contains 1000 news segments with various content, there are 13,388 speech segments in the database, each speech segment is about 10s and the total length is about 30 h. In this paper, we select 10 speech segments with different speech contents from 17 people, and performed a variety of content preserving operations (CPOs), for example amplitude adjustment, noise addition, re-quantization, resampling, MP3, etc., get a total of 3060 speeches for training. In the experimental analysis stage, randomly select 1000 speeches in the THCHS-30 speech database for evaluation.

The experimental hardware platform is: Intel(R) Celeron(R) CPU N2940, 1.83GHz, and 8GB of memory. The software environment is: Windows 10, MATLAB R2016a, JetBrains PyCharm Community Edition 2019.2.4 × 64.

5.1 Performance analysis of deep hash coding model

5.1.1 Model accuracy analysis

In the encrypted speech retrieval system, the feature extraction of speech data and the generation process of deep hash coding play a crucial role, which lays the foundation for

subsequent efficient retrieval tasks. Based on the basic CNN structure, this paper designs an end-to-end deep hash coding model suitable for speech data, which is used to learn the deep hashing features of speech data. In network model, the setting of the number of nodes in the hash layer determines the length of final hash code, which represents the feature representation ability of the model. In order to get the best performance of the network model, the number of hash layer nodes in the designed model is tested. In this paper, experiments are conducted with the number of nodes in the hash layer being 64/128/256/384/512, and then evaluate their test accuracy. Figure 5 shows the test accuracy curve of the proposed CNN model.

As can be seen from Fig. 5, when the number of nodes in the hash layer is 512, the test accuracy of the model is the highest, but when the batch reaches 30, the accuracy is almost the same as that under the 384 nodes, this means the 384 nodes can meet the binary representation of the input speech data. Besides, the too long hash code will lead to the decrease of retrieval efficiency, and the too short hash code will affect the feature representation of data, resulting in a reduction in the discriminability between hash codes. From the analysis of model accuracy, the proposed method believes that the 384-node model structure performs well and can be used to generate compact deep hash codes. The proposed CNN network model has achieved good test accuracy in the different node settings of the hash layer, which shows that the network fits the speech data well, without overfitting and underfitting. As shown in Fig. 5, with the increase of the number of nodes, the test accuracy of the model is also increasing, and it is infinitely close to 1, that because the more nodes, the more nonlinear the model will be, the more it can fit the input data to a greater extent, it is easier to get a higher accuracy. Therefore, the deep hash coding model based on CNN structure designed in this paper has good performance, and can achieve better accuracy in speech data, which is helpful for deep hash coding operation of speech data.

5.1.2 The mean average precision analysis

In addition, the experiment adopts mean Average Precision (mAP) to further measure the performance of the network model. The experiment uses models set by different nodes to calculate the *AP* value of the speech after various content preserving operations, where *AP*

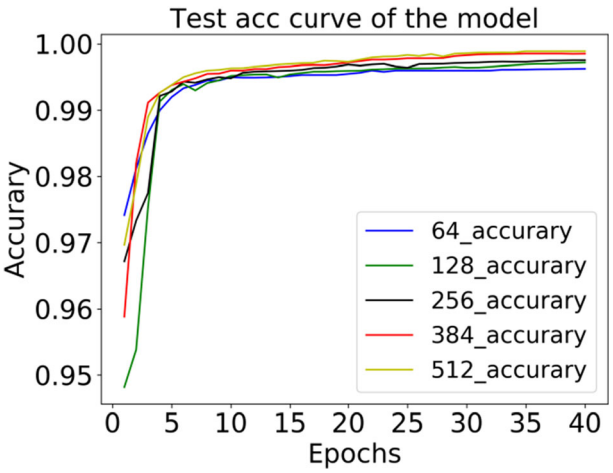


Fig. 5 Test accuracy curve of the proposed CNN model

value is the average precision, and then the *mAP* is calculated. The calculation formula are shown in Eqs. (11) and (12):

$$AP(q) = \frac{\sum_{i=1}^{n}(P(i) \times rel(i))}{L}, rel(i) \in \{0, 1\} \tag{11}$$

$$MAP = \frac{\sum_{q=1}^{Q} AP(q)}{Q} \tag{12}$$

where n represents the total number of speeches, L represents the number of speeches related to the query speech, Q indicates the number of query speeches, and $rel(i)$ refers to the correlation between the speech at position i in the search result and the query speech, if it is related to the query speech, it is 1, otherwise it is 0.

Table 2 shows the *mAP* obtained by the model under different hash code settings.

It can be seen from Table 2, the mAP increases with the increase of hash encoding length, which is because the longer hash encoding can provide more sufficient semantic feature representation for the input speech data. When the hash code length is 384/512, the obtained mAP has reached more than 0.95, and there is no significant difference between the mAP of 384 nodes and 512 nodes, it means that under 384/512 nodes, the network has basically reached the optimal performance and has better query performance for input speech data. Combined with the above experimental analysis of the model accuracy, and balanced the retrieval efficiency of the system, the number of nodes in the hash layer is set to 384, to obtain the 384 bit binary hash code for the subsequent related experiments.

5.2 Deep hash binary code performance analysis

5.2.1 Discriminant analysis

The proposed method measures the similarity of the constructed hash sequences by calculating the bit error rate (BER, also called the normalized Hamming distance) between speeches of different contents. The BER calculation formula as shown in Eq. (13):

$$BER(b_{x_1}, b_{x_2}) = \sum_{i=1}^{N}(|b_{x_1}(i) - b_{x_2}(i)|)/N \tag{13}$$

where b_{x_1} and b_{x_2} represent the deep hashing sequence of the speech files x_1 and x_2, respectively, and N is the length of the hash sequence.

In an ideal condition, the BER between different content speeches will present a normal distribution. The experiment randomly selects 1000 speeches with different contents in THCHS-30 speech database to test. By pairwise matching the hash codes of 1000 speeches,

Table 2 *mAP* under different hash code settings

Hash code length	64	128	256	384	512
mAP	0.8909	0.9281	0.9450	0.9631	0.9696

the experiment obtained a total of 1000 × 999/2 = 499,500 BER data. The BER normal probability distribution and BER frequency statistical histogram of the matching results are shown in Figs. 6 and 7.

As shown in Figs. 6 and 7, the probability distribution of the BER value of different content speech segments basically overlaps with the standard normal distribution curve, indicating that the hash sequence generated by the proposed deep hashing construction method is approximately normal distribution, and has good randomness and collision resistance. The BER statistical results obtained in this paper follow a normal distribution with mean $\mu = 0.4787$, standard deviation $\sigma = 0.0324$, and a range of 0.3203–0.6276. In other words, the minimum difference between the two test speeches is 0.3203, which can fully meet the retrieval needs.

In addition, the FAR (False Accept Rate) under different thresholds can further reflect the discriminability of the algorithm. For two speeches, calculate the BER between them, if it is less than the set threshold, they are determined to be the same content speech, otherwise, they are determined to be different content speech; and the FAR means that the proportion of different speech as the same content speech. Generally, the larger the set threshold, the larger the FAR, indicating the lower discriminability; on the contrary, the smaller the FAR, the better the discriminability.

The FAR calculation formula is defined as shown in Eq. (14).

$$FAR(\tau) = \int_{-\infty}^{\tau} f(x|\mu,\sigma) = \frac{1}{\sqrt{2\pi}\sigma}\int_{-\infty}^{\tau} e^{\frac{-(x-\mu)^2}{2\sigma^2}}dx \tag{14}$$

where τ is the hash sequence matching threshold, x is the bit error rate (BER), μ refers to BER mean, and σ refers to BER standard deviation.

Compare the proposed deep hashing method with the most representative perceptual hashing method in recent years. The comparison results of FAR values under different thresholds are shown in Table 3.

It can be seen from Table 3, under different thresholds, the FAR of the proposed method is far less than Ref. [18, 25, 27, 31], when $\tau = 0.16$, the FAR of the proposed method is 3.92 ×

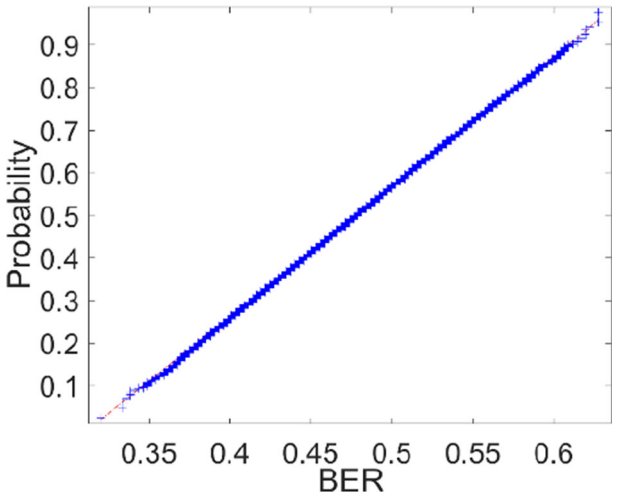


Fig. 6 BER normal probability distribution

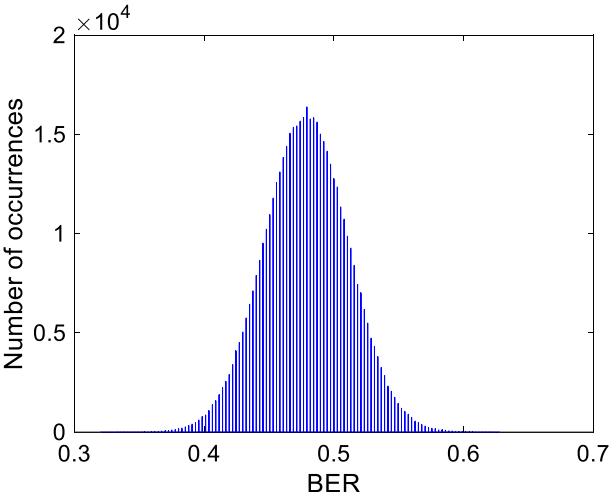


Fig. 7 BER frequency statistical histogram

10^{-23}, which means that the number of false judgments in 1023 speeches is only 3.9, which means the proposed method has great discriminability. However, the FAR of the proposed method is slightly higher than that of Ref. [35], which is because Ref. [35] adopts the feature selection step and speech denoising algorithm to improve the discriminability of feature hash. Although the proposed method has some gaps compared with Ref. [35], it also meets the needs of retrieval and can get better retrieval results. In summary, the proposed method has well discriminability and great representation ability for speech data.

5.2.2 Robustness analysis

Robustness means that after the original speech is processed by CPO such as re-quantization, resampling, MP3 compression, etc., the generated hash sequence is still consistent with the hash sequence generated by the original speech, and does not affect its subsequent retrieval. In order to test the robustness of the proposed method, randomly select 1000 speeches with different contents to perform the CPO as shown in Table 4, then a total of 4000 speeches are obtained. Table 4 shows the mean BER after various CPOs.

It can be seen from Table 4 that under amplitude reduction operation, the BER of the proposed method is smaller than Ref. [8, 25, 32, 37], which is slightly larger than Ref. [35], while in the

Table 3 Comparison of FAR under different thresholds

τ	Proposed	Ref. [25]	Ref. [37]	Ref. [32]	Ref. [8]	Ref. [35]
0.02	8.40×10^{-46}	4.18×10^{-19}	4.27×10^{-26}	4.78×10^{-38}	8.44×10^{-42}	1.94×10^{-56}
0.04	4.53×10^{-42}	1.14×10^{-17}	4.25×10^{-24}	4.22×10^{-35}	1.60×10^{-38}	5.27×10^{-52}
0.06	1.67×10^{-38}	2.70×10^{-16}	3.48×10^{-22}	2.81×10^{-32}	2.20×10^{-35}	9.32×10^{-48}
0.08	4.23×10^{-35}	5.59×10^{-15}	2.35×10^{-20}	1.40×10^{-29}	2.20×10^{-32}	1.07×10^{-43}
0.10	7.32×10^{-32}	1.00×10^{-13}	1.30×10^{-18}	5.25×10^{-27}	1.59×10^{-29}	8.01×10^{-40}
0.12	8.67×10^{-29}	1.56×10^{-12}	5.96×10^{-17}	1.48×10^{-24}	8.40×10^{-27}	3.89×10^{-36}
0.14	7.04×10^{-26}	2.12×10^{-11}	2.25×10^{-15}	3.15×10^{-22}	3.21×10^{-24}	1.23×10^{-32}
0.16	3.92×10^{-23}	2.51×10^{-10}	6.97×10^{-14}	5.02×10^{-20}	8.90×10^{-22}	2.53×10^{-29}

amplitude increase operation, the BER of the proposed method is smaller than Ref. [25, 32, 37], but slightly lower than Ref. [8, 35], in the MP3 compression operation, the BER of the proposed method is also smaller than Ref. [25, 32, 35], under resampling operation, the BER of the proposed method is also better than Ref. [25, 35, 37]. The robustness of the proposed method is slightly lower than the comparison literature under some specific operations, the main reason is that the spectrogram is the feature in time-frequency domain, and individual details will inevitably be ignored during the extraction process, and the multifractal characteristic adopted in Ref. [37] and the syllable-level perceptual feature adopted in Ref. [8] have relatively complex model structure, which is more robust than time-frequency domain hashing. The Ref. [35] added the denoising algorithm, feature denoising is helpful to improve the robustness. Although the robustness of the proposed method is not optimal under MP3 compression and resampling operation, it is also the same order of magnitude as the comparative literature, which already conforms the robustness requirements, so the proposed method has great robustness.

5.3 Retrieval performance analysis

In this paper, recall rate (R) and precision rate (P) are used to evaluate the retrieval performance of the system. Where the recall rate (R) represents the proportion of the amount of related queries detected to the total amount of related queries in the retrieval system. The calculation formula as shown in Eq. (15):

$$R = \frac{TP}{TP + FN} \times 100\% \tag{15}$$

where TP indicates the number of speech fragments retrieved correctly, FN refers to the number of unretrieved speeches related to the query, that is, the number of "undetected", $TP + FN$ is the total number of speeches related to the query in the database.

The precision rate (P) is also called the retrieval accuracy, which indicates the proportion of the correct number of samples in all returned results. The calculation formula as shown in Eq. (16):

$$P = \frac{TP}{TP + FP} \times 100\% \tag{16}$$

where FP indicates the number of speech fragments retrieved incorrectly, so the $TP + FP$ is the total number of returned speeches.

When retrieving the query, it is necessary to set a threshold to match the query hash sequence b_q with the hash sequence b_x in the deep hash index table. Set the similarity threshold to T, $0 < T < 0.5$, if the Hamming distance $D(b_q, b_x) < T$ between two hash sequences, the match is successful and return the retrieval result. Therefore, the selection of similarity threshold T will directly affect the matching result. In the above discriminability experiment, the minimum BER is 0.3203, in the robustness experiment, the maximum BER is 0.2344, therefore, in order to get the best query results, the similarity threshold is determined in the range of 0.2344–0.3203, and

Table 4 Comparison of mean BER of CPOs

CPOs	Proposed	Ref. [25]	Ref. [37]	Ref. [32]	Ref. [8]	Ref. [35]
Amplitude reduction/−3 dB	0.0032	0.0157	0.0139	0.0038	0.0042	0.0014
Amplitude increase/+3 dB	0.0101	0.0557	0.0476	0.0160	0.0039	0.0052
MP3 compression	0.0047	0.0067	0.0038	0.0090	0.0016	0.1928
Resampling/8–16 kbps	0.0092	0.0959	0.0693	0.0033	0.0026	0.0283

set to $T = 0.27$. Through Eqs. (15) and (16), the recall rate and precision rate of speech after various CPOs are tested. Table 5 represents the experimental results.

As can be seen from Table 5, the proposed method still has high recall rate and precision rate after various CPOs, both of them have reached 100%, which is better than the method proposed in Ref. [25] and Ref. [8], slightly better than Ref. [35], and is equivalent to the performance in Ref. [32, 37]. Therefore, the method in this paper still has better query effect under various CPOs.

Figure 8 shows the P-R curve of the proposed method and Ref. [8, 25, 32, 35, 37]. As shown in Fig. 8, the area enclosed by the P-R curve of the proposed method is larger than Ref. [8, 25, 32, 35, 37], which shows that the retrieval performance of this paper is the best. The area formed by the P-R curve of Ref. [8, 25, 32, 37] is much smaller than the proposed method, in Ref. [35], although the precision rate is higher than the proposed method when the recall rate is 0.97, when the recall rate is greater than 0.97, the precision rate drops rapidly, which is lower than the proposed method, so the area enclosed by P-R curve is also smaller than the proposed method. In summary, the proposed method has well retrieval performance compared to the comparative literature.

In the retrieval experiment of the query speech, randomly select the 300th speech as the query speech, the hash sequence of the query speech is constructed by the proposed method, and then perform hash matching in the deep hash index table, the result is shown in Fig. 9.

As shown in Fig. 9, except for the query speech, the BER of the remaining 999 speeches are greater than the set threshold of 0.27, and the matching fails, while the 300th speech is matched successful.

5.4 Retrieval efficiency analysis

Finally, experiment tests the retrieval efficiency of this method, randomly select 10,000 speeches in the THCHS-30 speech database for evaluation, calculate the average retrieval time of the proposed method, and compare with the method in Ref. [8, 25, 32, 35, 37], the result is shown in Table 6.

It can be seen from Table 6, compared with Ref. [8, 25, 32, 37], the retrieval efficiency of the proposed method is higher, and the retrieval performance is great. The retrieval efficiency of this paper is about 8.2 times of Ref. [25], 7.8 times of Ref. [37], 8.6 times of Ref. [8], and

Table 5 Comparison of recall and precision under different CPOs

CPOs		MP3 compression	Resampling/8–16 kbps	Amplitude reduction/−3 dB	Amplitude increase/+3 dB
Recall rate R	Proposed	100%	100%	100%	100%
	Ref. [25]	98%	96%	100%	100%
	Ref. [37]	100%	100%	100%	100%
	Ref. [32]	100%	100%	100%	100%
	Ref. [8]	96%	98%	–	–
	Ref. [35]	100%	100%	100%	100%
Precision rate P	Proposed	100%	100%	100%	100%
	Ref. [25]	97%	97%	98%	99%
	Ref. [37]	100%	100%	100%	100%
	Ref. [32]	100%	100%	100%	100%
	Ref. [8]	96%	97%	–	–
	Ref. [35]	98%	100%	100%	100%

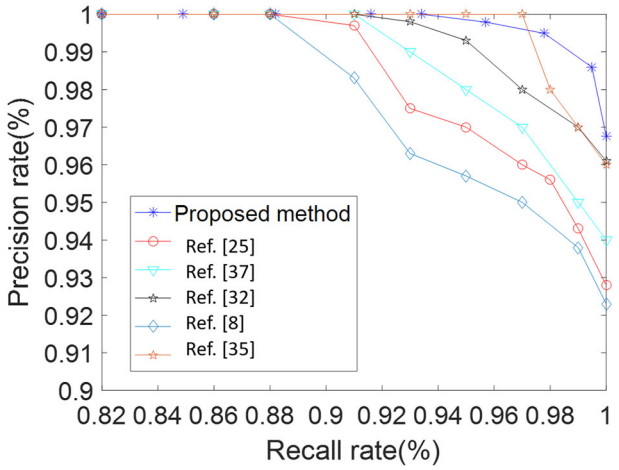


Fig. 8 P-R curve comparison results

slightly better than Ref. [32]. It shows that the deep hash binary code obtained in this paper has high quality and strong feature representation ability, which greatly promotes the retrieval efficiency. While the Ref. [8, 25, 32, 37] is to extract manual features in advance, and then construct perceptual hash sequence through hash function, the construction process is slightly complicated and its features are not very meaningful, so it will have an adverse effect on retrieval efficiency of the system. In contrast, the retrieval efficiency of this paper is lower than Ref. [35] a little, because K-means clustering method is used in Ref. [35] to classify the speech features, and it also compresses the length of the long binary hash code to shorten the matching length, but the process of constructing hash sequence is more complex than proposed method. Therefore, although the speech length used in this paper is longer than Ref. [8, 25, 32, 35, 37], it still achieves higher retrieval efficiency and retrieval performance.

In summary, the proposed CNN-based deep hash coding model has high accuracy, and achieves great mAP under different length hash node settings, which is suitable for the extraction of advanced semantic features and generation of hash coding in encrypted speech

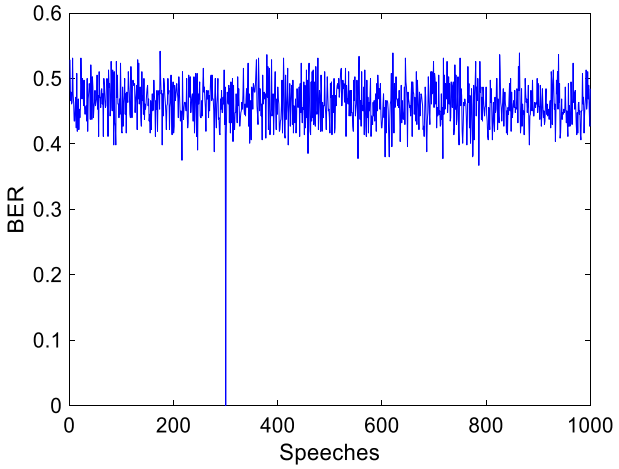


Fig. 9 The matching result of query speech

Table 6 Retrieval efficiency comparison

Methods	Frequency (GHz)	Speech length (s)	Average time (s)
Ref. [25]	–	6	3.9760
Ref. [37]	3.20	4	3.7937
Ref. [32]	2.90	4	0.4932
Ref. [8]	3.20	4	4.2032
Ref. [35]	2.50	4	0.1814
Proposed	1.83	10	0.4854

retrieval system. At the same time, compared with other representative perceptual hash codes, the deep hash codes constructed by the proposed method have better discriminability and robustness, and still have high recall rate, precision rate and efficiency under various CPOs.

6 Conclusions and future work

In order to solve the problems of poor feature semantics in the feature extraction process, low retrieval precision and efficiency of existing methods, referring to the advantages of deep hashing methods in the fields of image processing and speaker recognition, we have proposed a content-based encrypted speech retrieval scheme with deep hashing. The proposed method uses the designed deep hash coding model based on CNN structure to mine the deep semantic features and learn hash function, which breaks through the limitations of manual feature in traditional feature extraction process; In addition, a compact binary hash code is directly generated by using the learned hash function, which lays a solid foundation for subsequent retrieval steps and solves the problems of low retrieval precision and efficiency of traditional retrieval method. At the same time, introducing the BN algorithm in model training, which speeds up the convergence of the model and improves the robustness and generalization ability of the network model. The experimental results show that, compared with the existing method, the deep hash code generated by the proposed method has better discriminability and robustness, as well as higher retrieval accuracy and efficiency.

The disadvantage of the proposed method is that the deep hash index table is used for traversal search, resulting in relatively low retrieval efficiency. The next research is to introduce “classification” method, in order to reduce the candidate set of the query speech to further improve the retrieval precision and efficiency.

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